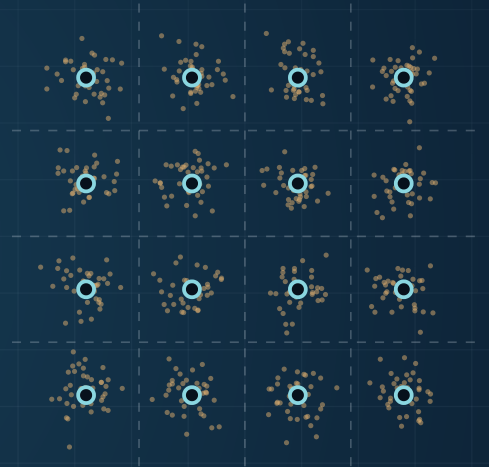
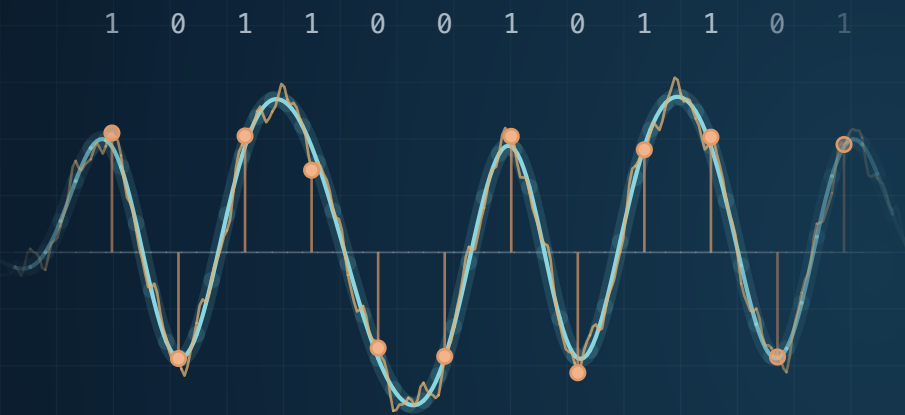




SAMPLING · DETECTION · MODULATION · CODING

Digital Communications

Formula and Notation Reference



Contents

*The conventions used throughout the course, every formula it establishes, and every symbol it defines.
Nothing here is derived; the derivations are in the lecture notes.*

1 Conventions

Energy and power, noise, the Q function, sinc, logarithms, and the two expressions every error probability comes from.

2 Summary of formulas

Everything the course establishes, in the order it establishes it.

3 Notation

Every symbol the course defines.

How to read it

Part 1 states the conventions, the two expressions every error probability comes from, and the transform pair Chapters 1 and 2 use. Part 2 is the summary of formulas, in the order the course establishes them. Part 3 defines every symbol. Nothing here is derived: where a result needs an argument, the argument is in the lecture notes chapter named beside it.

Conventions

These hold everywhere in the course, without local variation. Half the factor-of-two errors in this subject come from mixing two of them in one line, so where a result depends on one it is restated at that point rather than assumed.

CONVENTION	STATEMENT
Energy and power	Normalised: the resistance is taken as 1Ω , so instantaneous power is $ x(t) ^2$ and energy is its integral. Signal energy is E , energy per bit E_b , energy per symbol E_s , and $E_s = (\log_2 M) E_b$.
Noise	White and Gaussian with two-sided power spectral density $S_n(f) = N_0/2$. Projected onto an orthonormal basis it gives independent components, each $\mathcal{N}(0, N_0/2)$.
The Q function	$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$. It is the tail of a standard normal, so $Q(0) = \frac{1}{2}$ and $Q(-x) = 1 - Q(x)$.
sinc	Normalised: $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$, with zeros at every non-zero integer. Stated again wherever it is used.
Logarithms	log without a base means base two, and every entropy and every bit count is in bits. Decibels are $10 \log_{10}$ of a power or energy ratio.
Signals	$s_i(t)$ is transmitted, $r(t)$ is received, $\{\psi_k(t)\}$ is the orthonormal basis, and boldface $\mathbf{s}_i$ is the point in signal space.
Priors	Symbols are equally likely unless a problem says otherwise, in which case the MAP form is used and the prior term written out.

THE TWO EXPRESSIONS THE WHOLE COURSE RESTS ON

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

The first is exact for two signal points and is where every error probability in Chapters 2, 4 and 5 comes from. The second is its nearest-neighbour extension to M points, and the 2 in the denominator is the noise variance per dimension being $N_0/2$.

THE FOURIER TRANSFORM PAIR, AS CHAPTERS 1 AND 2 USE IT

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$

$$x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft} df$$

Written in hertz rather than in rad/s, so no factor of 2π appears in front of either integral. Sampling and the Nyquist criterion are both stated in this form.

PART 2

Summary of formulas

Everything the course establishes, in the order it establishes it. Nothing here is derived — where a result needs an argument, the argument is in the chapter named beside it.

A.1 From an analog signal to bits — Chapter 1

SAMPLING AND RECONSTRUCTION

$f_s > 2W$ (the sampling theorem, W the highest frequency present)

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Below the Nyquist rate the replicas of the spectrum overlap and the overlap cannot be undone. $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$ throughout this course.

UNIFORM QUANTIZATION

$$\Delta = \frac{2V}{L} = \frac{2V}{2^n}, \quad |e| \leq \frac{\Delta}{2}$$

$$\sigma_q^2 = \frac{\Delta^2}{12}, \quad \text{SQNR}|_{\text{dB}} = 6.02n + 1.76$$

The 1.76 assumes a full-scale sinusoid. The formula is an approximation at small n . At three bits it gives 19.82 dB where the measured value is 19.09 dB, and the gap halves with each extra bit.

RESULT	STATEMENT	CHAPTER
Companding	μ -law and A-law compress before quantizing so that the signal-to-noise ratio is flat across the range	1
PCM rate	$R = nf_s$ bits a second for n bits a sample	1

A.2 Baseband transmission — Chapter 2

THE MATCHED FILTER

$$h(t) = s(T - t), \quad \left(\frac{S}{N}\right)_{\max} = \frac{2E}{N_0}$$

The matched filter maximizes the signal-to-noise ratio at the sampling instant. The maximum depends on pulse energy, not pulse shape.

BINARY ERROR PROBABILITY AT BASEBAND

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \text{ (antipodal),} \quad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right) \text{ (on-off, orthogonal)}$$

NYQUIST CRITERION FOR ZERO INTERSYMBOL INTERFERENCE

$$\sum_k P\left(f + \frac{k}{T}\right) = T \quad \text{for all } f$$

$$B = \frac{1 + \alpha}{2T}, \quad 0 \leq \alpha \leq 1$$

The raised cosine meets the criterion for every roll-off α . At $\alpha = 0$ the bandwidth is the Nyquist minimum $1/2T$ and the pulse decays slowly. At $\alpha = 1$ it is twice that and the pulse decays quickly.

A.3 Signal space — Chapter 3

A SIGNAL AS A POINT

$$s_i(t) = \sum_{k=1}^N s_{ik} \psi_k(t), \quad s_{ik} = \int_0^T s_i(t) \psi_k(t) dt$$

$$E_i = \|\mathbf{s}_i\|^2 = \sum_{k=1}^N s_{ik}^2, \quad d_{ij} = \|\mathbf{s}_i - \mathbf{s}_j\|$$

The basis $\{\psi_k\}$ is orthonormal and comes from the Gram–Schmidt procedure applied to the signal set. Energy becomes a squared length and difference becomes a distance, which is the move the rest of the course rests on.

A.4 The optimal receiver — Chapter 4

THE DECISION RULE

$$\hat{s} = \arg \min_i \left\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \right\} \quad (\text{MAP})$$

$$\hat{s} = \arg \min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad (\text{ML, equal priors})$$

With equal priors the rule is minimum distance. The equivalent correlation form is to maximise $\mathbf{r} \cdot \mathbf{s}_i - E_i/2$.

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right)$$

$$P_e \leq \frac{1}{M} \sum_k \sum_{j \neq k} Q\left(\sqrt{\frac{d_{kj}^2}{2N_0}}\right), \quad P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

The first result is exact for two points. The second is the union bound. The third is its nearest-neighbor form. Here, $N_{\min}$ is the average number of nearest neighbors and can be noninteger. The loose bound needs only M and $d_{\min}$.

A.5 Digital modulation — Chapter 5

SCHEME	$d_{\min}^2$	$N_{\min}$
BPSK	$4E_b$	1
BFSK, BASK	$2E_b$	1
M -PSK	$4E_s \sin^2(\pi/M)$	2
M -PAM	$12E_s/(M^2 - 1)$	$2(M - 1)/M$
M -QAM (square)	$6E_s/(M - 1)$	3 at $M = 16$, 3.5 at $M = 64$
M -FSK	$2E_s$ for every pair	$M - 1$

READING THE TABLE

$$P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right), \quad E_s = (\log_2 M)E_b$$

Convert once between symbol and bit energy. BFSK and BASK need 3.01 dB more than BPSK for the same error probability. Doubling M costs approximately 6 dB in PAM. Moving from 4-PSK to 8-PSK costs 5.33 dB. At the same energy, 16-QAM gains 9.29 dB over 16-PAM.

A.6 Information theory — Chapter 6

INFORMATION AND ENTROPY

$$I(s_k) = -\log_2 p_k, \quad H(S) = -\sum_{k=1}^K p_k \log_2 p_k$$

$$0 \leq H(S) \leq \log_2 K, \quad H(S^n) = n H(S)$$

The upper bound is reached when all K symbols are equally likely. The extension result holds because the source is memoryless.

SOURCE CODING

$$\bar{L} = \sum_{k=1}^K p_k l_k, \quad \eta = \frac{H(S)}{\bar{L}} \leq 1$$

$$\sum_{k=1}^K 2^{-l_k} \leq 1, \quad H(S) \leq \bar{L} < H(S) + 1$$

The Kraft inequality is necessary but not sufficient for a prefix code. Over the n -th extension the upper bound becomes $H(S) + 1/n$, so $L_n/n \rightarrow H(S)$.

HUFFMAN CODES

$$\sigma^2 = \sum_{k=1}^K p_k (l_k - \bar{L})^2$$

Huffman is optimal in average length and is not unique. Placing a merged symbol as high as possible on a tie gives the minimum-variance code among the optimal ones.

PART 3

Notation

Every symbol the course defines, with the chapter that defines it. A symbol is never reused for a second meaning.

f_s, T_s

Sampling rate in hertz and the interval between samples, $T_s = 1/f_s$.

W

The highest frequency present in a message. The sampling theorem asks for $f_s > 2W$.

$\text{sinc}(x)$

Normalised: $\sin(\pi x)/(\pi x)$, with zeros at every non-zero integer.

Δ, L, n

Quantizer step, number of levels and bits a sample, with $L = 2^n$ and $\Delta = 2V/L$ over a range $\pm V$.

σ_q^2

Quantization noise power, $\Delta^2/12$ for a uniform quantizer.

T, R

Symbol interval and the rate that follows from it. A bit interval is written T_b and a bit rate R_b .

$p(t), h(t)$

The transmitted pulse and the impulse response of the receive filter. The matched filter is $h(t) = p(T - t)$.

α

Roll-off of a raised-cosine pulse, from 0 to 1. Bandwidth is $(1 + \alpha)/2T$.

$\psi_k(t)$

The k -th orthonormal basis function, from the Gram–Schmidt procedure applied to the signal set.

$s_i(t), \mathbf{s}_i$

The i -th transmitted waveform, and the point in signal space whose coordinates are its projections onto the basis.

E, E_i

Signal energy, normalised so that no resistance appears: $E_i = \|\mathbf{s}_i\|^2$.

$d_{ij}, d_{\min}$

Distance between two signal points, and the smallest such distance in a constellation.

$r(t), \mathbf{r}$

The received waveform and the observation vector it produces, $\mathbf{r} = \mathbf{s}_i + \mathbf{n}$.

N_0

Noise power spectral density parameter. The two-sided density is $N_0/2$ watts per hertz, and each noise component is $\mathcal{N}(0, N_0/2)$.

$Q(x)$

The Gaussian tail: the probability that a standard normal variable exceeds x . Equal to $\frac{1}{2} \text{erfc}(x/\sqrt{2})$.

M, N

The number of signals in the set, and the number of basis functions they need. M signals carry $\log_2 M$ bits a symbol.

$N_{\min}$

The average number of signal points at the minimum distance. An average, so not usually a whole number.

P_b, P_e, P_s

Bit error probability, error probability in general, and symbol error probability.

E_b, E_s

Energy per information bit and energy per transmitted symbol, with $E_s = (\log_2 M)E_b$.

f_c

Carrier frequency. Every scheme in Chapter 5 changes the amplitude, the phase or the frequency of $\cos(2\pi f_c t)$.

S, K, p_k

The source alphabet, how many symbols it has, and the probability of the k -th.

$I(s_k)$

Self-information of one symbol, $-\log_2 p_k$ bits.

$H(S)$

Entropy: the average information a symbol carries, in bits a symbol. Bounded by 0 and $\log_2 K$.

$\bar{L}, l_k$

Average codeword length and the length of the codeword for symbol k .

η

Coding efficiency, $H(S)/\bar{L}$. Never above one.

σ^2

Variance of the codeword length about $\bar{L}$. Separates two Huffman codes that share an average.

$p(y_k | x_j)$

Transition probability of a discrete memoryless channel. Collected into the channel matrix, whose every row sums to one.

$H(X | Y)$

Conditional entropy: the uncertainty still remaining about the input after the output has been seen.

$I(X; Y)$

Mutual information, $H(X) - H(X | Y)$: what the channel actually delivered, in bits per use. Symmetric and never negative.

C

Channel capacity, the largest $I(X; Y)$ over all input distributions. For the binary symmetric channel, $1 - H(p)$.

$H(p)$

The binary entropy function, $-p \log_2 p - (1 - p) \log_2 (1 - p)$. It measures a two-symbol source and also the noise a binary symmetric channel adds.